# C. Scoring Subsequences

time limit per test 2.5 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: binary search, greedy, math, two pointers | https://codeforces.com/problemset/problem/1794/C

The *score* of a sequence $[s_1, s_2, \ldots, s_d]$ is defined as $(s_1 \cdot s_2 \cdot \ldots \cdot s_d)/(d!)$, where $d! = 1 \cdot 2 \cdot \ldots \cdot d$. In particular, the score of an empty sequence is 1.

For a sequence $[s_1, s_2, \ldots, s_d]$, let $m$ be the maximum score among all its subsequences. Its *cost* is defined as the maximum length of a subsequence with a score of $m$.

You are given a **non-decreasing** sequence $[a_1, a_2, \ldots, a_n]$ of integers of length $n$. In other words, the condition $a_1 \le a_2 \le \ldots \le a_n$ is satisfied. For each $k = 1, 2, \ldots, n$, find the cost of the sequence $[a_1, a_2, \ldots, a_k]$.

A sequence $x$ is a subsequence of a sequence $y$ if $x$ can be obtained from $y$ by deletion of several (possibly, zero or all) elements.

## Input

Each test contains multiple test cases. The first line contains the number of test cases $t$ ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains an integer $n$ ($1 \le n \le 10^5$) — the length of the given sequence.

The second line of each test case contains $n$ integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le n$) — the given sequence. It is guaranteed that its elements are in non-decreasing order.

It is guaranteed that the sum of $n$ over all test cases does not exceed $5 \cdot 10^5$.

## Output

For each test case, output $n$ integers — the costs of sequences $[a_1, a_2, \ldots, a_k]$ in ascending order of $k$.

## Example

**Input**

```
3
3
1 2 3
2
1 1
5
5 5 5 5 5
```

**Output**

```
1 1 2
1 1
1 2 3 4 5
```

## Note

In the first test case:

- The maximum score among the subsequences of [1] is 1. The subsequences [1] and [] (the empty sequence) are the only ones with this score. Thus, the cost of [1] is 1.
- The maximum score among the subsequences of [1, 2] is 2. The only subsequence with this score is [2]. Thus, the cost of [1, 2] is 1.
- The maximum score among the subsequences of [1, 2, 3] is 3. The subsequences [2, 3] and [3] are the only ones with this score. Thus, the cost of [1, 2, 3] is 2.

Therefore, the answer to this case is 1 1 2, which are the costs of [1], [1, 2] and [1, 2, 3] in this order.